

Newton's Method and Extension

↓
Secant
method

↓
false position
method

① newton's method :

⇒ formula :

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, n \geq 1$$

⇒ need to solve : p_0 will given in Q

Q1 : 1. Let $f(x) = x^2 - 6$ and $p_0 = 1$. Use Newton's method to find p_2 .

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

→ find $f'(x)$:

$$f(x) = x^2 - 6$$

$$f'(x) = 2x$$

→ find p_2 :

$$p_0 = 1$$

$$p_1 = 1 - \frac{(1)^2 - 6}{2(1)} = 3.5$$

$$P_2 = 3.5 - \frac{(3.5)^2 - 6}{2(3.5)} = 2.6071428$$

Q 2. Let $f(x) = -x^3 - \cos x$ and $p_0 = -1$. Use Newton's method to find p_2 . Could $p_0 = 0$ be used?

$$P_n = P_{n-1} - \frac{f(P_{n-1})}{f'(P_{n-1})}$$

→ find $f'(x)$ &

$$f(x) = -x^3 - \cos x$$

$$f'(x) = -3x^2 - (-\sin x) = -3x^2 + \sin x$$

$$\text{note} \Rightarrow \frac{d}{dx} \cos x = -\sin x$$

→ find P_2 &

$$P_0 = -1$$

$$P_1 = P_0 - \frac{-(P_0)^3 - \cos(P_0)}{-3(P_0)^2 + \sin(P_0)} = -0.88033289$$

$$P_2 = P_1 - \frac{-(P_1)^3 - \cos(P_1)}{-3(P_1)^2 + \sin(P_1)} = -0.8656841$$

→ If we can use $P_0 = 0$?

No, we can't due to $f'(P_0) \neq 0$

Q50 5. Use Newton's method to find solutions accurate to within 10^{-4} for the following problems.

a. $x^3 - 2x^2 - 5 = 0$, $[1, 4]$

$$P_n = P_{n-1} - \frac{f(P_{n-1})}{f'(P_{n-1})}$$

→ find $f(x)$ &

$$f(x) = x^3 - 2x^2 - 5$$

$$f'(x) = 3x^2 - 4x$$

→ find $P > 10^{-4}$ &

for P_0 you can take any number between $[1, 4]$

or you can take mean value :

$$= P_0 = \frac{1+4}{2} = 2.5 \approx 2$$

$$\Rightarrow P_0 = 2$$

$$P_1 = 2 - \frac{(2)^3 - 2(2)^2 - 5}{3(2)^2 - 4(2)} = 3.25$$

$$P_2 = P_1 - \frac{(P_1)^3 - 2(P_1)^2 - 5}{3(P_1)^2 - 4(P_1)} = 2.8110367$$

$$P_3 = P_2 - \frac{(P_2)^3 - 2(P_2)^2 - 5}{3(P_2)^2 - 4(P_2)} = 2.6979894$$

$$P_4 = P_3 - \frac{(P_3)^3 - 2(P_3)^2 - 5}{3(P_3)^2 - 4(P_3)} = \underline{2.6906771}$$

$$P_5 = P_4 - \frac{(P_4)^3 - 2(P_3)^2 - 5}{3(P_3)^2 - 4(P_3)} = \boxed{2.6906474} \rightarrow |P_5 - P_4| = 0.00002 < 10^{-4}$$

Q5/d ° d. $x - 0.8 - 0.2 \sin x = 0, [0, \pi/2]$

① find $f(x)$ °

$$\rightarrow f(x) = x - 0.8 - 0.2 \sin x$$

$$f'(x) = 1 - 0.2 \cos x$$

note : $\frac{d}{dx} \sin x = \cos x$

② find $p > 10^{-4}$

$$P_0 = \frac{0 + \frac{\pi}{2}}{2} = \frac{\pi}{4}$$

$$P_1 = \frac{\pi}{4} - \frac{(\frac{\pi}{4}) - 0.8 - 0.2 \sin(\frac{\pi}{4})}{1 - 0.2 \cos(\frac{\pi}{4})} = 0.9671208$$

$$P_2 = P_1 - \frac{(P_1) - 0.8 - 0.2 \sin(P_1)}{1 - 0.2 \cos(P_1)} = 0.9643346$$

$$P_3 = P_2 - \frac{(P_2) - 0.8 - 0.2 \sin(P_2)}{1 - 0.2 \cos(P_2)} = \boxed{0.9643338} \rightarrow |P_3 - P_2| = 0.0000008 < 10^{-4}$$

② Secant method

Formula

$$P_n = P_{n-1} - \frac{f(P_{n-1})(P_{n-1} - P_{n-2})}{f(P_{n-1}) - f(P_{n-2})}$$

→ Always start w/ P_2

→ Should give P_0 and P_1 to find P_2

Q3. Let $f(x) = x^2 - 6$. With $p_0 = 3$ and $p_1 = 2$, find p_3 .

a. Use the Secant method.

$$P_n = P_{n-1} - \frac{f(P_{n-1})(P_{n-1} - P_{n-2})}{f(P_{n-1}) - f(P_{n-2})}$$

→ find P_3

was given $P_0 = 3$, $P_1 = 2$

$$P_2 = P_1 - \frac{f(P_1)(P_1 - P_0)}{f(P_1) - f(P_0)} = 2.4$$

$$\Rightarrow f(P_0) = (3)^2 - 6 = 3$$

$$\Rightarrow f(P_1) = (2)^2 - 6 = -2$$

$$P_3 = P_2 - \frac{f(P_2)(P_2 - P_1)}{f(P_2) - f(P_1)} = 2.45454545$$

$$\Rightarrow f(P_2) = (2.4)^2 - 6 = -0.24$$

Q4% 4. Let $f(x) = -x^3 - \cos x$. With $p_0 = -1$ and $p_1 = 0$, find p_3 .

a. Use the Secant method.

$\Rightarrow$ Find p_3 % $p_0 = -1$, $p_1 = 0$

$$p_2 = p_1 - \frac{f(p_1)(p_1 - p_0)}{f(p_1) - f(p_0)} = -0.6850734$$

$$\Rightarrow f(p_0) = -(-1)^3 - \cos(-1) = 0.4596976$$

$$\Rightarrow f(p_1) = -(0)^2 - \cos(0) = -1$$

$$p_3 = p_2 - \frac{f(p_2)(p_2 - p_1)}{f(p_2) - f(p_1)} = \boxed{-1.2519648}$$

$$\Rightarrow f(p_2) = -(p_2)^3 - \cos(p_2) = -0.4528014$$

Q5% 5. Use Newton's method to find solutions accurate to within 10^{-4} for the following problems.

c. $x - \cos x = 0$, $[0, \pi/2]$

We can assume $p_0 = 0$ $p_1 = \pi/2$

$$p_2 = p_1 - \frac{f(p_1)(p_1 - p_0)}{f(p_1) - f(p_0)} = 0.6110154$$

$$\Rightarrow f(p_0) = 0 - \cos(0) = -1$$

$$\Rightarrow f(p_1) = \pi/2 - \cos(\pi/2) = \pi/2$$

$$P_3 = P_2 - \frac{f(P_2)(P_2 - P_1)}{f(P_2) - f(P_1)} = 0.7232695$$

$$\Rightarrow f(P_2) = (P_2) - \cos(P_2) = -0.2080505$$

$$P_4 = P_3 - \frac{f(P_3)(P_3 - P_2)}{f(P_3) - f(P_2)} = \underline{0.7395670}$$

$$\Rightarrow f(P_3) = (P_3) - \cos(P_3) = -0.0263763$$

$$P_5 = P_4 - \frac{f(P_4)(P_4 - P_3)}{f(P_4) - f(P_3)} = \underline{0.7395203} \rightarrow 4 \text{ digit are similar}$$

$$|P_5 - P_4| = 0.00004 < 10^{-4}$$

$$\Rightarrow f(P_4) = (P_4) - \cos(P_4) = 0.0008065$$

d. $x - 0.8 - 0.2 \sin x = 0, \quad [0, \pi/2]$

$$P_0 = 0 \quad \text{and} \quad P_1 = \pi/2$$

$$\Rightarrow P_2 = P_1 - \frac{f(P_1)(P_1 - P_0)}{f(P_1) - f(P_0)} = 0.9167206$$

$$\Rightarrow f(P_0) = 0 - 0.8 - 0.2 \sin(0) = -0.8$$

$$\Rightarrow f(P_1) = \pi/2 - 0.8 - 0.2 \sin(\pi/2) = 0.570796$$

$$P_3 = P_2 - \frac{f(P_2)(P_2 - P_1)}{f(P_2) - f(P_1)} = 0.9615658$$

Remark 3

How to know you

Not going wrong

in calculation by

the similarity

in outcomes.

$$f(p_2) - f(p_1)$$

$$\Rightarrow f(p_2) = (p_2) - 0.8 - 0.2 \sin(p_2) = -0.0420161$$

$$p_4 = p_3 - \frac{f(p_3)(p_3 - p_2)}{f(p_3) - f(p_2)} = \underline{0.9643449}$$

$$f(p_3) = (p_3) - 0.8 - 0.2 \sin(p_3) = -0.0024519$$

$$p_5 = p_4 - \frac{f(p_4)(p_4 - p_3)}{f(p_4) - f(p_3)} = \underline{0.9643338} \quad \rightarrow 4 \text{ digit are similar}$$

$$|p_5 - p_4| = 0.00001 < 10^{-4}$$

$$\Rightarrow f(p_4) = 0.0000976$$

.....

③ False Position Method

$$p_n = p_{n-1} - \frac{f(p_{n-1})(p_{n-1} - p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}$$

$\Rightarrow$ in false position method we will use BiSection method and Secant method. How?

① check $f(p_0)$ and $f(p_1)$ is must in Opposite sign's

② find p_2 by using formula that used in Secant method

③ find $f(p_2)$ depends on sign replace Interval

Q38

3. Let $f(x) = x^2 - 6$. With $p_0 = 3$ and $p_1 = 2$, find p_3 .

b. Use the method of False Position.

① check $f(p_0)$ and $f(p_1)$ is must in Opposite signs

$$\Rightarrow f(p_0) = (3)^2 - 6 = 3, +ve$$

$$\Rightarrow f(p_1) = (2)^2 - 6 = -2, -ve$$

② find p_2 by using formula that used in Secant method

$$\Rightarrow p_2 = p_1 - \frac{f(p_1)(p_1 - p_0)}{f(p_1) - f(p_0)}$$

$$= 2 - \frac{(-2)(2-3)}{(-2) - (3)} = 2.4$$



③ find $f(p_2)$ depends on sign replace interval

$$f(p_2) = (2.4)^2 - 6 = -0.24 \Rightarrow [3, 2.4]$$

$$p_3 = p_2 - \frac{f(p_2)(p_2 - p_1)}{f(p_2) - f(p_1)} = 2.4 - \frac{(-0.24)(2.4 - 3)}{-0.24 - 3} = \boxed{2.4444444}$$

Q98

9. Repeat Exercise 5 using the method of False Position.

5. Use Newton's method to find solutions accurate to within 10^{-4} for the following problems.

d. $x - 0.8 - 0.2 \sin x = 0, \quad [0, \pi/2]$

$$f(p_0) = (0) - 0.8 - 0.2 \sin(0) = -0.8, -ve$$

$$f(p_1) = (\pi/2) - 0.8 - 0.2 \sin(\pi/2) = 0.5707963, +ve$$



$$p_2 = p_1 - \frac{f(p_1)(p_1 - p_0)}{f(p_1) - f(p_0)} = 0.9167204$$

$$\Rightarrow f(p_2) = (p_2) - 0.8 - 0.2 \sin(p_2) = -0.0420016, -ve \Rightarrow [0, -0.9615512]$$

$$p_3 = p_2 - \frac{f(p_2)(p_2 - p_1)}{f(p_2) - f(p_1)} = 0.9615512$$

$$\Rightarrow f(p_3) = (p_3) - 0.8 - 0.2 \sin(p_3) = -0.0024648, -ve \Rightarrow [0, -0.0024648]$$

$$p_4 = p_3 - \frac{f(p_3)(p_3 - p_2)}{f(p_3) - f(p_2)} = 0.9643460$$

$$\Rightarrow f(p_4) = p_4 - 0.8 - 0.2 \sin(p_4) = 0.0000107, +ve \Rightarrow [0.0000107, -0.0024648]$$

$$p_5 = p_4 - \frac{f(p_4)(p_4 - p_3)}{f(p_4) - f(p_3)} = 0.9643339 \rightarrow 4 \text{ digit are similar}$$

$$|p_5 - p_4| = 0.00001 < 10^{-4}$$